

27/07/2026 HW:-

Q) Trace Quick sort as shown in the class for the following array:- (i) 85, 76, 99, 13, 45, 103, -5, 26, 31

(ii) 85, 76, 99, 13, 45, 103, -5, 26, 31

(i) 85, 76, 99, 13, 45, 103, -5, 26, 31

85	76	31	13	45	103	-5	26	99
i	i	i	i	i	<u>i</u>		j	j
					↑		↑	

85	76	31	13	45	26	-5	108	99
8	7	3	1	4	2	-	1	9
5	6	<u>1</u>	3	5	6	5	08	9

8, j crossed each other.

qs(a, 0, 5);

qs(a, 7, 8);

-5	76	31	13	45	26	85	103	99
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-5	76	31	13	45	26	85	103	99
----	----	----	----	----	----	----	-----	----

i
j

j j j j j

i

qs(a, 6, 6) x

qs(a, 7, 8).

qs(a, 0, 0)

qs(a, 1, 5)

85	99	103
----	----	-----

-5	76	31	13	45	26
----	----	----	----	----	----

j j j j j
↑

-5	26	31	13	45	76
----	----	----	----	----	----

85	99	103
----	----	-----

qs(a, 10, 4)

qs(a, 5, 5) x

-5	26	31	13	45
----	----	----	----	----

76

85	99	103
----	----	-----

j j j
↑

-5	13	31	26	45
----	----	----	----	----

76

85	99	103
----	----	-----

-5

j j j j j
j j j j j

-5	13	26	31	45	76	85	99	103
----	----	----	----	----	----	----	----	-----

n = 9, first = 0, last = 8, pivot = 0.

i	j	a[i]	a[j]	a[pivot]
0	8	85	31	85
2	8	99	31	85
5	7	103	26	85
7	6	103	-5	85
8	6	85	-5	85

i > j swap pivot & j

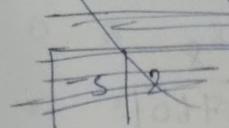
99 ↔ 31

103 ↔ 26

85

85

final Array



~~first~~ first = 1, last = 5, pivot = 1

i	j	a[i]	a[j]	a[pivot]	
1	5	76	26	76	Initial
6	5	85	26	76	76 $\leftrightarrow$ 21
1	5	76	26	76	26 $\leftrightarrow$ 13

first = 1, last = 4, pivot = 1.

i	j	a[i]	a[j]	a[pivot]	
1	4	26	45	26	Initial
2	3	31	13	26	31 $\leftrightarrow$ 13
3	2	31	13	26	i > j
1	2	26	13	26	a[pivot] $\leftrightarrow$ 13

first = 3, last = 4, pivot = 3

i	j	a[i]	a[j]	a[pivot]	
3	4	31	45	31	Initial
4	3	45	31	31	i > j
3	3	31	31	31	same pivot

first = 7, last = 8, pivot = 7

i	j	a[i]	a[j]	a[pivot]	
7	8	103	99	103	Initial
8	8	99	99	103	i = j
9	8	no element	99	103	swap 103 $\leftrightarrow$ 99

∴ final array =

-5	13	26	31	45	76	85	99	103
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(ii) 350, 790, 400, 200, 100, 96, 900, 890, 760

given,

0	1	2	3	4	5	6	7	8
350	790	400	200	100	96	900	890	760
i					j			

350	96	400	200	100	790	900	890	760
i				j				

350	96	100	200	400	790	900	890	760
i					j			

$i > j \rightarrow \text{swap (pivot \& j)}$

200	96	100	350	400	790	900	890	760
-----	----	-----	-----	-----	-----	-----	-----	-----

qs(a, 0, 2)

qs(a, 4, 8)

200	96	100	350	400	790	900	890	760
i		j						

qs(a, 0, 1)

qs(a, 2, 2) x

qs(a, 4, 4) x

qs(a, 5, 8)

100	96	200	350	400	790	900	890	760
i	j							

96	100	200	350	400	790	760	890	900
					i		j	

760	790	890	900
-----	-----	-----	-----

890	900
-----	-----

--	--

890	900
-----	-----

96	100	200	350	400	760	790	890	900
----	-----	-----	-----	-----	-----	-----	-----	-----

i	j	a[i]	a[j]	a[pivot]	
0	8	350	760	350	Initial
1	5	790	96	350	swap 790 ↔ 96
2	4	400	100	350	swap 400 ↔ 100
4	3	400	200	350	i > j

~~200 96 100~~

200	96	100
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i	j	a[i]	a[j]	a[pivot]	
0	2	200	100	200	Initial
-	2	200	100	200	i > j

q(a, 0, 1)

q(a, 2, 2) x

i	j	a[i]	a[j]	a[pivot]	
4	8	400	760	400	Initial.
5	8	790	760	790	Initial.
7	6	890	900	790	Initial.
7	8	890	900	890	Initial.
5	4	790	400	400	i > j

qs(a, 4, 4) x

qs(a, 5, 8)

96	100	200	350	400	790	900	890	760
----	-----	-----	-----	-----	-----	-----	-----	-----

i	j	a[i]	a[j]	a[pivot]	
6	8	900	760	790	swap 900 ↔ 760
7	6	890	760	790	i > j

96	100	200	350	400	760
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790	900	890
-----	-----	-----

i > j
swap pivot & j

∴ final array =

96	100	200	350	400	760	790	890	900
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pass = 3

3	8	12	19
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3	8	12
---	---	----

3	8
---	---

∴ final array =

3	8	12	19	31	46	54
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HW ÷ 30/04/2026

Do the quick sort trailing for the above values.
~~3, 8, 12, 31, 19, 46, 54~~ 54, 12, 19, 3, 8, 46, 31

given,

54	12	19	3	8	46	31
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$i > j$
swap j & pivot

~~| | | | | | | |
|----|----|----|---|---|----|----|
| 31 | 12 | 19 | 3 | 8 | 46 | 54 |
|----|----|----|---|---|----|----|~~

~~$i > j$ swap j & pivot~~
 $qs(a, 0, 5) \checkmark$
 $qs(a, 6, 6) \times$

31	12	19	3	8	46	54
----	----	----	---	---	----	----

31	12	19	8	3	46	54
----	----	----	---	---	----	----

$i > j \Rightarrow$ swap j & pivot
 $qs(a, 0, 3) \checkmark$
 $qs(a, 4, 5) \checkmark$

3	12	19	8	31	46	54
---	----	----	---	----	----	----

3	12	8	19	31	46	54
---	----	---	----	----	----	----

~~$i > j \Rightarrow$ swap pivot & j~~
 ~~$qs(a, 1, 1) \times$~~
 ~~$qs(a, 2, 3) \times$~~

3	8	12	19	31	46	54
---	---	----	----	----	----	----

3	8	12	19	31	46	54
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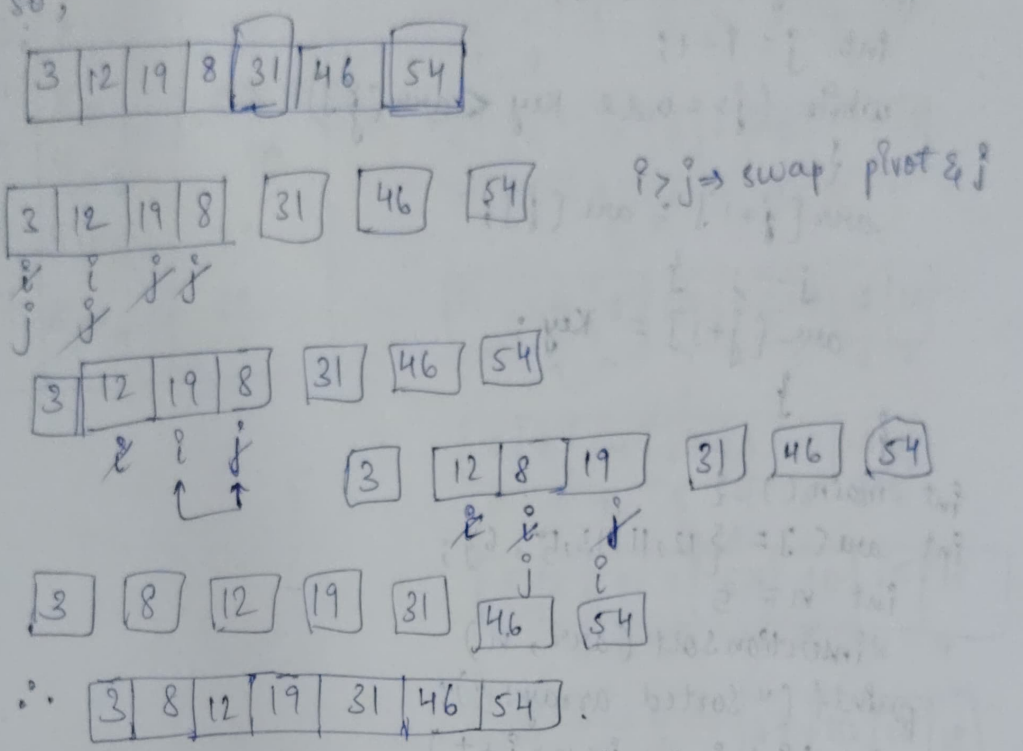
~~swap pivot & j~~

∴ final array =

3	8	12	19	31	46	54
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$n = 7$; $first = 0$; $last = 6$
 $i \quad j \quad a[i] \quad a[j] \quad a[pivot]$
 $0 \quad 6 \quad 54 \quad 31 \quad 54$ Initial
 $- \quad 6 \quad - \quad 31 \quad 54$ $i > j$ (swap pivot & j)

So,



i	j	$a[i]$	$a[j]$	$a[pivot]$	
0	5	31	46	31	Initial
3	4	3	8	31	$i < j \Rightarrow$ swap i & j
0	5	31	46	31	
5	4	46	3	31	$i > j \Rightarrow$ swap pivot & j
0	3	3	8	3	Initial & j
1	0	12	3	3	$i > j \Rightarrow$ swap j & pivot
1	3	12	8	12	Initial
2	3	19	8	12	swap i & j ($\because i < j$)
1	3	12	19	12	Initial
3	2	19	8	12	$i > j \Rightarrow$ swap pivot & j

$\therefore$ final array =

